

I · WHY · RESEARCHERS AND LABS · 0.2

FarmNotary

A research notary for published runs.
The simulation stays off-chain.

github.com/Dooders/FarmNotary



I · WHY · AUDIENCE

You, if a stranger has to check the run.

THIS IS FOR YOU

- You publish simulation or computational-experiment artifacts.
- A reviewer should not have to trust a laptop folder, a zip, or a Drive link.
- You want language for what the bytes allow — and a hard stop after that.

NOT THE PITCH

- Not “put the simulation on a chain.”
- Not a score, rank, or lab reputation token.
- Not a claim that a notarized run is scientifically correct.

I · WHY · THE GAP

Published runs still live on someone's laptop.

FETCH

A reviewer cannot fetch the official artifacts without asking you for a folder.

REHASH

They cannot rehash those bytes against a record that is not the zip itself.

DATE

They cannot see when that exact file was published, except by trusting the email.

A narrower job than “the science is right,” and a broader one than `sha256sum`.

Matching bytes is the start of the sentence.

A HASH TOOL

These files match this digest.

That is the entire claim.

A RESEARCH NOTARY

- Which files were admitted, by a recorded allowlist.
- How many were left out — without printing their names.
- Whether the design was stamped before the artifacts existed.
- Whether those exact bytes existed by time T.
- What you may claim. What you may not.

I · WHY · WHAT IT IS

The official record, dated, without running a chain.

01 · WRITE

A canonical `manifest.json` — code identity, config, artifact hashes.

02 · PIN

Optional remote pin so the CID is a citation, not a laptop.

03 · STAMP

OpenTimestamps calendars batch the content hash into Bitcoin. No contract, no keys, no gas.

Immutability is not correctness. A manifest can perfectly notarize a wrong result.

Build what a stranger can check.

IN

- Canonical run and campaign manifests
- Publish profiles and a privacy denylist
- Optional remote pin (IPFS pinning service)
- Keyless Bitcoin timestamp (OpenTimestamps)
- Claim card, paper-pack appendix, static index
- GitHub Action and a small Python API

OUT

- Scientific correctness
- Preventing the file drawer
- Publishing ballots or individual choices
- Scores, rankings, protocol identity
- Operating calendars, contracts, or pin services
- Running the simulation in the EVM

Nothing is hashed until you say what may leave.

STAYS PRIVATE

Voter or agent choice. Paths containing ballot, vote, voter, individual_choice, or private — even if a glob would admit them.

OFFICIAL RECORD

Config, code identity, aggregate metrics, winner allocations. Named profiles so labs do not invent the allowlist.

01 PROFILE

```
farm-notary manifest --run-dir path/to/run --profile consensus
```

Profiles: consensus, rl-sweep, evolution-run. Unmatched files increment a count; their names are never printed.

Three commands, then a card.

01 HASH

```
farm-notary manifest --run-dir path/to/run --profile consensus --config config.json
```

02 VERIFY

```
farm-notary verify --run-dir path/to/run
```

03 STAMP

```
farm-notary anchor --run-dir path/to/run --backend ots --pin-remote pinata
```

Hours later: `farm-notary upgrade --run-dir path/to/run` completes the Bitcoin attestation. Pending calendars are not L0.

Reviewers read the card, not the exit code.

FARM-NOTARY VERIFY

CLAIM CARD

```
level: none — no earned ladder level
next:  L0 — these bytes existed by time T; Bitcoin headers not verified by this tool (missing: Bitcoin attestation)
• tamper-evident record           — pass
• existed by time T               — pending
• pre-specified design           — missing
• bitwise reproducible (scoped)  — 6/6, ignored: *.mp4; byte-identical on x86-64 Linux in a pinned environment
• not claimed: scientific correctness
```

Missing is not failure.

01 FETCH

```
ipfs get <cid> # or obtain the run directory another way
```

02 READ

```
farm-notary verify --run-dir <dir>
```

03 OPT IN

```
farm-notary verify --run-dir <dir> --verify-derived
```

Exit 0 means no attempted check failed – not that every claim was earned, and not that the science is right. Step 03 only if you trust the recorded commands.

Earn a rung. Do not decorate one.

NONE	No Bitcoin attestation yet, or the record does not rehash.	Not a failed experiment.
L0	These bytes existed by time T. Proof commits to the content hash and carries a Bitcoin-height attestation.	Not header verification. Use <code>ots verify</code> .
L1	Re-execution is specified: command, git SHA, environment fingerprint.	Not a completed re-run.
L2	Seed not grindable after the plan is anchored.	Not scientific correctness. Needs a live or fixture beacon compare.
L3	A Sigstore keyless signature is on the reproduction receipt. Identity is recorded, not proven independent.	Not independently reproduced.

Stamp the plan before the artifacts exist.

01 PLAN

```
farm-notary precommit --config config.json --command "python run.py --out {run_dir}" --backend ots
```

02 BIND

```
farm-notary manifest --run-dir path/to/run --profile consensus --precommit precommit.json
```

A timestamp taken after the result is known is not pre-registration.

FarmNotary will still stamp the favorable run. It will not call that a pre-specified design.

Re-run the seed. Scope the sentence.

DEMONSTRATED

8/8

100 trials, 300 voters, 8 candidates, seed 0. Summary statistics
recompute from `trials.csv`.

The only sentence the tool may emit: *byte-identical on x86-64 Linux
in a pinned environment.*

NOT A CLAIM

- Cross-hardware bitwise identity.
- Linux ARM smoke cell: 6/10. macOS ARM: 2/10.
- A classified byte-diff (path, timestamp, float print, video encoder) is **not a science failure**.

The figure is a parent record, not a folder.

01	SWEEP	<code>farm-notary campaign --name consensus-sweep --run-dir runs/seed-0 --run-dir runs/seed-1 --out sweep/</code>
02	APPENDIX	<code>farm-notary paper-pack --campaign sweep/ --out sweep/appendix.md</code>
03	INDEX	<code>farm-notary index --registry docs/registry.md --campaign sweep/</code>

CID	content-addressed run
CONTENT HASH	canonical body, stamps excluded
BITCOIN ATTESTATION	block N, or pending
ALLOWLIST / UNMATCHED	policy is part of the claim
READER LADDER	— do not cite Ln from the PDF

Free to stamp. Impossible to unpublish.

MACHINE

Python $\geq$ 3.9. Core is stdlib-only: manifest, verify, IPFS client, dry-run.

LIVE STAMP

`[ots]` extra. Public calendars. No key, no gas, Bitcoin-backed. EAS stays experimental.

DEFAULT

CLI `anchor` and `precommit` are dry-run. Nothing is hashed until an allowlist is declared. A remote pin has no retract.

Drop the Action in. Fail the job if the card fails.

DOODERS/FARMNOTARY@DEV

PHASE: NOTARIZE

```
- uses: dooders/FarmNotary@dev
  with:
    phase: notarize
    run-dir: results
    profile: consensus
    pin-remote: pinata
    backend: ots
```

Recommended: `phase: precommit` at job start, `phase: notarize` after the run. Pin `@dev` or a commit until v0.2.0 is tagged.

An anchor proves existence, not correctness.

REFUSED

- No “verified result” badge. No treating a printed Ln as a finding.
- No scoreboard, trusted-lab registry, or protocol token.
- No inventory of unpublished runs.

STILL TRUE

- Cherry-picking is out of scope; mislabeling a post-hoc stamp is not.
- The author’s own reproduction is bitwise (scoped), not independently reproduced.
- Cross-hardware bitwise identity is not a claim.

Install the 0.2 line. Hash one run.

01 INSTALL

```
pip install "farm-notary[ots] @ git+https://github.com/Dooders/FarmNotary.git@dev"
```

02 HASH

```
farm-notary manifest --run-dir path/to/run --profile consensus
```

03 READ

```
farm-notary verify --run-dir path/to/run
```

github.com/Dooders/FarmNotary

Docs: [PRINCIPLES](#) · [CLAIMS](#) · [DESIGN](#). Apache-2.0. PyPI still serves 0.1.0 – install from git for this deck.